SOME IMPORTANT POLYPHAGOUS PESTS

- Aphid, Thrips, Whiteflies, Borer- *Helicoverpa armigera* , Red spider mite, Red hairy caterpillar, Locust

Aphids:

- Known also as greenfly, blackfly or plant lice, aphids are minute plant-feeding insects.
- Important natural enemies include the predatory ladybugs/ladybirds/ladybeetles, and lacewings.
- Aphids are tiny, pear-shaped, sap-sucking pests, appearing in the on your plants' tender new leaves. They leave behind a secretion that attracts ants and promotes mold growth.

Thrips:

- Thrips (order Thysanoptera) are minute (mostly 1 mm long or less), slender insects with fringed wings and unique asymmetrical mouthparts.
- Thrips can cause damage during feeding.
- Nymph and adult lacerates leaves from the under surface of the leaves and flower buds.
- As a result, white streaks appear on the infested leaves. Leaves show brown patches and get distorted, finally wither and drop down. Infested flowers do not open, flowers fade and drop down prematurely.

Red spider mite

- Larvae, nymphs and adults cause damage to the host plant by feeding on plant sap.
- They mainly occur on the underside of leaves where they pierce the cells and suck out the contents.
- The empty dead cells become yellow, and in many plants the damage can also be seen on the upper surface of leaves as small yellow dots.
- The destruction of cells results in reduced photosynthesis, increased transpiration and reduced plant growth.
- As damage increases, whole leaves turn yellow, and as more cell sap is removed, the leaf, and eventually the whole plant, may die.
- The nymphs and adults also produce webs, and plants can get completely covered with such webs in which the mites live.
- The webbing and spotting on the leaves affects the appearance of the crop

Red hairy caterpillar (*Amsacta moorei)*

- **Family:** Arctiidae
- **Order:** Lepidoptera.
- Body of caterpillars is covered with the numerous long hairs arising from the fleshly tubers.
- The entire abdomen is scarlet red there are black bands and dots on the abdomen.
- Feed on gregariously and as they grow older moult 6 times.
- During a severe attack, the caterpillars in bands destroying fields after field.

❖ **White grub (***Holotrichia consanguinea)*

- **Family:** Melolonthidae
- **Order:** Coleoptera.
- Damage done by grub in field crops and by adults in trees.
- The damage caused by grub is so much so that entire stand crop is destroyed thereby necessitating the resowing in field.
- **Designated as a 'National pest'.**
- The plant damaged by the grub gives a wilted appearance and finally dries out.
- The full-grown grubs move down deeper in the soil in search of moisture & for pupation.
- After pre monsoon rain/ first shower of monsoon the adult comes out from soil and cause damage to trees during night.
- Eggs are laid in singly in loose moist sandy or sandy loam soil on the onset of monsoon.
- Both grubs and adults are damaging.
- Grubs feed on underground plant parts of various crops.
- Older second instar and third instar grubs are more damaging.
- Due to concealed feeding white grubs generally remain unnoticed and at harvest a large number of tubers are found infested/damaged.

- White grubs are also found to feed on the roots of horticultural/ forest nurseries and some ornamental plants.
- Adult beetles feed on the foliage of many trees.
- There are three larval instars.
- Beetles emerge during monsoon.
- Lay eggs in soil.
- Eggs are laid singly in 2-7 installments.
- Each female lays 4-40 eggs in its life span.
- The pest passes winter as grub in earthen cells.
- Pupate in cells during April – May.

❖ Locust:

- **Family:** Acrididae
- **Order:** Orthoptera
- Locusts are the swarming phase of short-horned grasshoppers.
- They can breed rapidly under suitable conditions and subsequently become gregarious and migratory.
- Nymphs form bands and adults' swarms.
- They can travel great distances, rapidly stripping fields and greatly damaging crops.
- The origin and apparent extinction of certain species of locust (some of which were 150 mm in length) is unclear.
- Adult and nymphs cause the damage.
- Gregarious and voracious feeders.
- Eat up the entire vegetation.
- In spite of some expensive control measures, the damage caused to crops by locusts during 1926-31 cycle was estimated to 100 million.
- They also climb on walls, invade kitchens, storerooms thus causing nuisance.

- ➢ Eggs are laid in soil in egg pods.
- ➢ Each female can lay up to 11 egg pods, each pod containing up to 120 eggs.
- ➢ Before egg laying each female with the help of its ovipositor, bores a 5-10 cm deep hole into the loose soil.

(a) **Bombay locust (Pantanga succinct L.):** Mostly confined to Gujarat and Tamil Nadu. Its breed during monsoon in the Western Ghats and has only one brood in a year.

(b) **Migratory locust (Locusta migratoria):** Mainly occurs in Rajasthan and Gujarat. Its breed twice in a year: winter -spring breeding occurs in Pakistan and summer-monsoon breeding in Rajasthan and Gujarat. It may, have many broods a year.

(c) **Desert locust (Schistocera gregarea):** This is most destructive of all locust The adult of this species occurs **in two phases:**

(i) **Solitary phase (solitaria):** wherein the insects remain scattered

(ii) **Gregarious phase (gregarea):** wherein the insects congregate as nymphs and adults.

- ➢ **Life -history:**
  - ✓ Mating occurs between sexually mature yellow adults.
  - ✓ Egg -laying starts after 8-24 hours of mating, a single female can lay up to 500 eggs in about 5 egg -pods or capsule.
  - ✓ Female inserts her long ovipositor in sandy soil to bore a hole 2-4 inches deep.
  - ✓ The summer breeding occurs during June —September (monsoon) in this quiescent eggs are hatched out and nymphal period is about 12-15 days, within this period emerges in five nymphal instars.
  - ✓ Within the period of one-month nymphal stage emerges into adult.
- ➢ **Damage:** Both nymphs and adults cause damage
- ➢ **Host plants:** Mainly pearl millet, sorghum, maize but being Polyphagous nature, it's eat on all kind of vegetation except a few namely, aak, neem, datura, Jamun and sheesham.
- ➢ **Management**:
  - ✓ Eggs can be destroyed by deep summer ploughing or flooding.

- ✓ The nymphs are most vulnerable stage for management point of view.
- ✓ Because at second and third nymphal stage is key stage where pest population can be checked.
- ✓ A trench can be dug around the breeding field of locust so that when nymphs emerge, they can be buried under soil, or methyl parathion 2% can be dusted to kill the emerging nymphs.
- ✓ When infestation starts on field crops or vegetation around the field duts, methyl parathion 2% or Malathion 5% @ 25 kg/ha on field crops including field border and vegetation of non-cropped area.

❖ **Termites (**Odontotermes obesus,)

- **Family:** Termitidae
- **Order:** Isoptera.
- Also known **as white ants**.
- Termite lives in social colony in underground nets make earthen moulds (termatoria).

A. Reproductive castes:

(i) **Colonizing individuals:** Winged individuals of both sexes and are produced in large number during rainy season. After mating they cast off the wings and start a new colony.

(ii) **Queen:** Only perfectly developed female in the colony. It measures about 5.0-7.5 cm in length, and she is known as a phenomenon, **'Egg laying machine'**. It is laying one egg/second or 70,000-80,000 eggs in 24 hours. Queen fed by workers; choicest food lives in "royal chamber" which situated in center of nest (0.5 m depth).

(iii) **King:** It developed from unfertilized eggs. It lives with queen in royal chamber & smaller than queen and mates with queen from time to time.

(iv) **Complementary caste:** Short winged or wingless of both sexes lead a subterranean life. They replaced in place of queen & king after ultimately death of them.

(v) **Sterile caste:**

a. Workers

b. Soldier

- **Management:**